DA-Flow: Degradation-Aware Optical Flow Estimation with Diffusion Models

Jaewon Min, Jaeeun Lee, Yeji Choi, Paul Hyunbin Cho, Jin Hyeon Kim, Tae-Young Lee et al. | KAIST AI

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Research Questions

- Can optical flow be accurately estimated from severely corrupted video inputs with blur, noise, and compression artifacts?
- How can image restoration diffusion models be adapted to encode temporal correspondences for optical flow estimation?
- What intermediate representations from diffusion models best capture geometric correspondence under severe degradation?

Core Idea

Existing optical flow methods fail catastrophically on corrupted videos because they're trained on clean data. The key insight is that image restoration diffusion models already encode degradation-aware representations that preserve geometric structure, but lack temporal awareness needed for motion estimation. The authors "lift" a pretrained image restoration diffusion model by adding cross-frame attention, enabling it to capture correspondences across time while maintaining degradation robustness. They then fuse these diffusion features with conventional CNN features in a hybrid architecture built on RAFT, allowing the model to benefit from both degradation-aware priors and fine-grained spatial details.

Key Formula

Full spatio-temporal attention reshape

$$\tilde{\mathbf{Q}}, \tilde{\mathbf{K}}, \tilde{\mathbf{V}} \in \mathbb{R}^{B \times (3FT) \times C}$$

B is batch size, F is number of frames, T is spatial tokens per frame, C is channel dimension. This reshapes modality streams from frame-independent processing to full spatio-temporal attention where each token attends to all spatial tokens across all frames and modalities, enabling inter-frame correspondence learning.

DA-Flow hybrid pipeline

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\mathcal{M}_{\theta} = \mathcal{U} \circ \mathcal{C} \circ (\texttt{Up}(\mathcal{D}_\phi), \mathcal{E})
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θ are flow network parameters, φ are diffusion model parameters. Up upsamples coarse diffusion features, D_φ is the lifted diffusion model, E is conventional CNN encoder. This combines degradation-aware diffusion features with fine-grained CNN features for correlation C and iterative updates U.

Key Figure

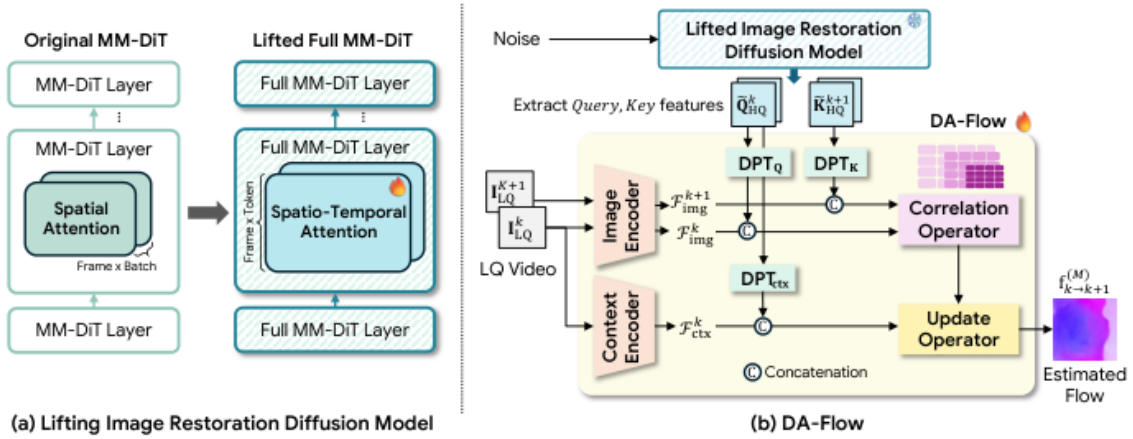

Fig. 2: Overall architecture of DA-Flow.

At inference, the model starts from pure noise and iteratively denoises the latent using the learned velocity field; the final restored image is then obtained by decoding the result with the VAE decoder.

4 Method

In this section, we present our approach to degradation-aware optical flow estimation. We begin by describing how a pretrained DiT-based image restoration model is lifted to the video domain through full spatio-temporal attention in Sec. 4.2. We then analyze the geometric correspondence encoded in the diffusion features across different layers, identifying which layers yield the most correspondence-ready representations in Sec. 4.3. Based on these findings, we introduce **DA-Flow**, our degradation-aware optical flow model that leverages the selected diffusion features to estimate reliable motion from corrupted inputs in Sec. 4.4.

4.1 Problem Formulation

We introduce a new task of **Degradation-Aware Optical Flow**, which aims to estimate accurate flow from corrupted videos. Let $\mathbf{V}_{\text{LQ}}$ and $\mathbf{V}_{\text{HQ}}$ denote a low-quality (LQ) video and its corresponding high-quality (HQ) video, respectively, each represented as a sequence of N RGB frames $\{\mathbf{I}^i\}_{i=1}^N$ with $\mathbf{I}^i \in \mathbb{R}^{3\times H\times W}$. Our goal is to learn a degradation-aware optical flow model $\mathcal{M}$ that reliably estimates motion from LQ inputs. For a pair of consecutive frames indexed by k and $k{+}1$, where $k \in \{1, \dots, N{-}1\}$, the model estimates:

$$\widehat{\mathbf{f}}_{k\to k+1} = \mathcal{M}(\mathbf{I}^k_{\text{LQ}},\ \mathbf{I}^{k+1}_{\text{LQ}}) \approx \mathbf{f}^*_{k\to k+1}, \tag{7}$$

where $\mathbf{f}^*_{k\to k+1}$ denotes the ground-truth flow between frames k and $k{+}1$. Among the three stages in Eq. 1, the feature encoder $\mathcal{E}$ is most directly affected by input degradation, as corrupted pixels lead to unreliable features that propagate

Figure 2: Shows the complete DA-Flow architecture including the lifting process from image restoration diffusion model to video

domain and the hybrid feature fusion strategy.

Limitations

- Requires pseudo ground-truth flow from pretrained models since real degraded video flow labels are unavailable
- Limited to specific degradation types used in training (Real-ESRGAN pipeline)
- Computational overhead from diffusion model inference during flow estimation
- Performance gains primarily demonstrated on synthetic degradations rather than real-world corrupted videos